Samuel Leblanc

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Education

Queen's University **2025 - 2027**
M.Sc. in Mathematics
Research project: Cluster algebras, generalized cluster categories, and representation theory
Supervisors: Ivan Dimitrov, Charles Paquette, and David Wehlau

Université de Sherbrooke **2022 - 2025**
B.Sc. in Mathematics
Mention of Excellence – Faculty Dean's List, Winter 2025

Research Interests

Representation theory of (finite-dimensional and associative) algebras, cluster algebras, homological algebra.

Papers

1. Brüstle, T., Desrochers, J., Leblanc S., *Generalized Rank via Minimal Subposet.* arXiv:2510.10837 (2025)
2. Leblanc, S., Rasolomanana, A., Armenta, M., *Hidden Activations Are Not Enough: A General Approach to Neural Network Predictions.* arXiv:2409.13163 (2024)
3. Armenta, M., Leblanc, S., *Batalin-Vilkovisky structure on Hochschild cohomology with coefficients in the dual algebra.* arXiv:1810.13023 (2018) (I joined as coauthor in 2025; revisions and extensions)

Student Papers

1. Baril, L., Leblanc, S., *Multiplicité de la représentation indicatrice.* Submitted to: Cahiers mathématiques de l'Université de Sherbrooke (2025)
   Supervisor: Thomas Brüstle
2. Leblanc, S., *Dégénérations des représentations de carquois de type $\mathbb{A}_3$ à deux puits.* To appear in: Cahiers mathématiques de l'Université de Sherbrooke (2024)
   Supervisors: Emily Cliff and Shiping Liu
3. Leblanc, S., *Transformations de cercles orientés tangents sur la sphère de Riemann.* Submitted to: Cahiers mathématiques de l'Université de Sherbrooke (2023)
   Supervisor: Jean-Philippe Burelle

Talks

1. *(Co)limit Computation and Its Application to Representation Theory* (January 26, 2026)
   Algebra & Geometry Seminar, Queen's University
2. *Generalized Rank in Generalized Linear Algebra* (November 12, 2025)
   Grad Seminar, Department of Mathematics and Statistics, Queen's University
3. *Analyse topologique de données* (February 13, 2025)
   Club mathématiques de l'Université de Sherbrooke
4. *La propagation avant en tant que matrice* (November 14, 2024)
   Club mathématiques de l'Université de Sherbrooke
5. *Visualisation de transformations sur la sphère de Riemann* (March 21, 2024)
   Club mathématiques de l'Université de Sherbrooke
6. *Théorie des représentations des réseaux de neurones* (October 5, 2023)
   Club mathématiques de l'Université de Sherbrooke

Posters

1. *Minimal Initial Subcategory.* Presented at Queen's Mathematics Summer School and Modern Methods in Representation Theory (2026)
2. *Multiplicity of the Interval Module.* Presented at AARMS-CMS Student Poster Session (CMS Summer Meeting) (2025)
   Collaborators: Laurianne Baril and Justin Desrochers. Supervisor: Thomas Brüstle.
3. *Kernel of the Rank Invariant* (with Justin Desrochers). Presented at Summer Research School, Applications of Representation Theory in Topological Data Analysis and Geometric Invariant Theory (2024)

Awards & Fellowships

Senator Frank Carrel Fellowship ($15,000) **2026 - 2027**
Queen's University

Best Poster Award (Minimal Initial Subcategory) **2026**
Queen's Mathematics Summer School

Teaching

Guest Lecturer

Real Analysis (MATH/MTHE 328) **2026**
Replaced the course instructor for one week. Delivered 3 hours of lectures.
Department of Mathematics and Statistics, Queen's University

Teaching Assistant

Differential and Integral Calculus (MATH 128) **2026**
Department of Mathematics and Statistics, Queen's University

Méthodes quantitatives en communication marketing (MQG301) **2025**
École de gestion, Université de Sherbrooke

Calcul vectoriel (MAT298) **2024**
Département de mathématiques, Université de Sherbrooke

Statistique appliquée à la gestion (MQG222) **2024**
École de gestion, Université de Sherbrooke

Grader

Differential and Integral Calculus, Real Analysis, Algebraic Structures, and Discrete Mathematics. **2024 - 2026**
Queen's University and *Université de Sherbrooke*

Service

Academic Coordinator, Queen's GMS **2026 - 2027**
Coorganizer, with Trevor Shillington, of the Graduate Student Seminar of the Department of Mathematics and Statistics.

Mathematics Tutor, 10th Grade **2023 - 2024**
Volunteering with Le Diplôme avant la Médaille

Selected Academic Activities

Conferences

Modern Methods in Representation Theory **2026**
Ottawa, ON.

Auslander Conference **2026**
Woods Hole, MA, USA.

Meeting on Representation Theory of Algebras **2024 & 2025**
Sherbrooke, QC.

Route 81 Conference **2025**
Kingston, ON.

Canadian Mathematical Society (CMS) Meeting **2025**
Québec, QC.

Summer Schools

Queen's Mathematics Summer School **2026**
Kingston, ON.

Summer Research School: Applications of Representation Theory in Topological Data Analysis and Geometric Invariant Theory **2024**
Montréal, QC.

Undergraduate Research

Summer Internship: Representation Theory of Quivers **2023**
Département de mathématiques, Université de Sherbrooke
Supervisors: Emily Cliff and Shiping Liu

Software

knowledgematrix GitHub
Implement neural networks and compute their knowledge matrices.

upperhalfplane GitHub
Visualize the action of $\mathrm{PSL}(2,\mathbb{R})$ on the upper half plane.

riemannsphere GitHub
Visualize the action of $\mathrm{PSL}(2,\mathbb{C})$ and $\mathrm{PSP}(4,\mathbb{R})$ on the Riemann sphere.
Supervisor: Jean-Philippe Burelle.

MatTalX GitHub
Extension to convert LaTeX commands into plain text.

Languages

French (native), English (advanced).